
Benchmarking and Optimizing Multimodal Structured Generation: The OracleGraph Dataset and PRISM Framework

Anonymous Author(s)

Affiliation

Address

email

Abstract

Structured generation remains a critical bottleneck for Multimodal Large Language Models (MLLMs). Current post-training paradigms struggle to resolve this: Supervised Fine-Tuning (SFT) hits a rigid ceiling, while scalar-reward Reinforcement Learning (e.g., DPO, GRPO, SimPO) suffers from a policy-level signal gap. Compressing long routing trajectories into sparse signals induces a short-completion bias and traps RL near the SFT baseline, whereas vanilla self-distillation actively regresses. To systematically quantify this joint visual-structural-relational reasoning, we introduce **OracleGraph**, a 10,000-page 2.5D historical document benchmark featuring a 12-dimensional diagnostic VQA suite and a structured-hallucination taxonomy. To overcome these optimization pathologies, we propose **PRISM**, a Policy-Reward Integrated Self-distillation framework. PRISM leverages multimodal information asymmetry via a Hindsight Teacher guided by a Diagnostic Graph Report (DGR). By synergizing Softmax Reward-Weighted Aggregation (preserving Graph-F1 rankings) with a Length-Normalized Auxiliary Preference Loss, PRISM explicitly neutralizes catastrophic truncation, yielding a +3.81pp Graph Reward gain and a +2.3pp schema-parsing improvement over SDPO ($p < 0.001$). While PRISM’s nominal +1.02pp mean gain over SFT is not statistically significant ($p = 0.12$), it drastically stabilizes optimization, achieving $2.4\times$ lower cross-seed variance. Crucially, on a cross-book out-of-distribution (OOD) probe, PRISM’s stability advantage amplifies to a $14\times$ variance reduction over SFT, confirming robust structural stabilization. Code and datasets are available at <https://anonymous.4open.science/r/PRISM2026>.

1 Introduction

Ancient Chinese documents – such as block-printed editions of *Honglouloumeng*, archival ledgers, and hand-copied manuscripts – pose a distinctive challenge for contemporary Multimodal Large Language Models (MLLMs). Unlike modern PDFs, each page is a richly annotated, non-linear surface: vertical main columns interleave with double-line interlinear commentary, top/bottom/side marginalia, fishtail marks, and cross-column fold lines, while vermilion seals and edit marks routinely overlap the text they modify. Parsing such a page into a strict, schema-compliant JSON graph of typed nodes and typed edges (e.g., READS_AFTER, ANNOTATES, OVERLAPS) is therefore not a cosmetic layout-analysis task, but a profound joint visual-structural-relational reasoning challenge. While reliable open-weight parsers for ancient Chinese corpora are heavily demanded by digital-humanities researchers, off-the-shelf 8B-class MLLMs fail almost completely in our measurements.

To bridge this structural reasoning gap, Supervised Fine-Tuning (SFT) is typically employed to install a schema prior. However, SFT fundamentally relies on behavioral cloning; without a mechanism to

explore and rank candidate parses, the model’s performance inevitably hits a rigid ceiling. The contemporary paradigm to break this ceiling is scalar-reward Reinforcement Learning (e.g., DPO, GRPO, SimPO). Yet, applying these algorithms to strict structured generation exposes fatal optimization vulnerabilities. Because complex JSON outputs often exceed thousands of tokens, compressing long sequence routing trajectories into a single sparse scalar reward strips away localized topological signals. Consequently, the policy suffers from a *policy-level signal gap*. Instead of learning fine-grained structural alignment, the policy actively resorts to a *short-completion bias*—truncating outputs and generating structurally thinner graphs to evade the zero-reward cliffs associated with partial parses.

To systematically quantify and overcome these optimization pathologies, we introduce two coupled pieces: **OracleGraph** and **PRISM**. OracleGraph is a comprehensive 10,000-page 2.5D historical-document benchmark featuring a typed-graph ontology, a fine-grained hallucination taxonomy, and a diagnostic Visual Question Answering (VQA) suite designed to expose structural blind spots. Built upon this benchmark, we propose PRISM (Policy-Reward Integrated Self-distillation for Multimodal structured extraction), a reference self-distillation framework tailored for long structured JSON generation. PRISM deliberately engineers multimodal information asymmetry via a Hindsight Teacher guided by a Diagnostic Graph Report (DGR). By synergizing softmax reward-weighted aggregation (which preserves Graph-F1 rankings) with a length-normalized auxiliary preference loss, PRISM explicitly neutralizes the short-completion bias and stabilizes the policy without relying on external reward models.

The contributions provided in the paper are summarized as follows:

- **Typed-Graph Ontology and Taxonomy:** We introduce the first 2.5D typed-graph ontology for ancient Chinese documents (13 node classes, 8 edge classes, 5-level occlusion z -index), enabling a 30+ category structured-hallucination taxonomy that is strictly finer-grained than standard single-metric evaluations like POPE or CHAIR.
- **The OracleGraph Benchmark:** We release a 10,000-page hand-annotated corpus ($\sim 275K$ typed annotations) coupled with a 12-dimensional diagnostic VQA suite. Our 8B domain-specific SFT stack dominates the structural subset of this benchmark, establishing the strongest open-weight sub-10B MLLM baseline for this domain.
- **Empirical RL Regime Diagnosis:** Under a rigorous, matched-rank ($r=32$) protocol with 3 seeds and $n=100$ paired bootstrap, we reveal that standard scalar-reward RL methods (DPO, GRPO, SimPO) fail to surpass the SFT ceiling ($p > 0.05$). They cluster within $\pm 1.5pp$ of the SFT baseline, driven by the identified short-completion bias and policy-level signal gap.
- **The PRISM Framework:** We propose a reward-aware self-distillation recipe that empirically breaks this bottleneck. PRISM yields a strict $+3.81pp$ Graph Reward gain over vanilla distillation ($p < 0.001$) and exhibits $2.4\times$ lower cross-seed variance than SFT, successfully stabilizing long-horizon structured generation.
- **Graph-Grounded Diagnostic Critique (DGR):** We demonstrate that the Teacher’s schema-aware critique (DGR) serves as a potent, reward-sensitive independent lever. Specifically, an auto-verbosity router (auto) and structural-statistics-only format (L2) empirically dominate non-DGR RL baselines on both mean reward and cross-seed stability.

2 Related Work

Our PRISM framework sits at the intersection of multimodal structured generation, reinforcement learning (RL), and knowledge distillation. We review the literature across these domains, highlighting gaps that necessitate our proposed algorithm.

2.1 Multimodal Structured Generation and Layout Analysis

Recent advancements in multimodal structured generation have shifted from multi-stage OCR pipelines to unified architectures that align textual features with high-resolution visual embeddings [Hu et al., 2024]. Recent works model layout analysis as end-to-end topological graph generation [Chen et al., 2025], or apply programmatic constraints to logits for strict structured decoding [Cesista, 2025]. Despite these improvements, Supervised Fine-Tuning (SFT) relies on behavioral cloning, struggling with robust spatial reasoning [Rizzoli et al., 2025]. Multimodal RL attempts to break the

87 SFT plateau using execution logic checks [Chen et al., 2026a] or perceptual verification [Wang et al.,
88 2025]. However, these nascent approaches rely on sparse scalar rewards or ad-hoc verifiers, limiting
89 efficacy on highly entangled 2.5D layouts. PRISM distinguishes itself by quantifying structural
90 boundaries via IsoTrain-100K and overcoming the SFT ceiling through a dense, reference-free
91 self-distillation paradigm.

92 2.2 Preference Optimization and the Credit Assignment Pathology

93 Applying scalar-reward algorithms (e.g., DPO, GRPO) [Ju et al., 2025, Rudner et al., 2021] to
94 strict structured generation exposes fatal vulnerabilities. While open-domain RL frequently exhibits
95 reward hacking via verbosity [Wang et al., 2026, MacDiarmid et al., 2025], complex JSON or graph
96 generation presents an unforgiving “zero-reward cliff” [Beigi et al., 2026, Zhou et al., 2025]. This
97 forces models into a *short-completion bias*—actively truncating outputs to minimize fatal syntax
98 errors.

99 This evasion is driven by the credit assignment pathology of outcome-supervised reward models
100 (ORM) [Lyu et al., 2025, Ding et al., 2026], where compressing routing decisions into a single scalar
101 strips localized schema signals. Process-supervised reward models (PRM) assign dense step-level
102 credit [Fei et al., 2025, Zhang et al., 2025a], but annotating intermediate 2.5D topological steps
103 remains prohibitively expensive. Reference-free methods bypass external models using length-
104 normalized margins to mathematically counteract the autoregressive length penalty and suppress
105 truncation biases [Qiyuan et al., 2026, Zou et al., 2026, Chen et al., 2026b, Liao et al., 2026, Gu et al.,
106 2025]. Nevertheless, these 1D sequence-level approaches remain blind to high-dimensional spatial
107 dependencies [Zhang et al., 2025b, Weng et al., 2025, Li et al., 2026a]. PRISM synthesizes these
108 insights, integrating length-normalized margins with asymmetric self-distillation to yield PRM-like
109 dense guidance without annotation overhead.

110 2.3 Self-Distillation and Information Asymmetry

111 Self-distillation iteratively improves models by aligning them with high-quality self-generations or an
112 advanced Teacher [Hübotter et al., 2026, Li et al., 2026c]. However, vanilla Self-Distillation Policy
113 Optimization (SDPO) struggles with complex structured tasks due to several inherent limitations.
114 First, it assumes information symmetry, which traps the policy in blind exploration. To overcome
115 this, PRISM draws on Learning Under Privileged Information (LUPI) [Li et al., 2026b, Yang et al.,
116 2026] by exposing the ground-truth JSON to the Teacher via a Diagnostic Graph Report (DGR). This
117 intentional information asymmetry provides optimal token-level supervision with globally unbiased
118 gradients, effectively rescuing the Student from zero-reward cliffs.

119 Beyond exploration bottlenecks, vanilla SDPO suffers from reward-agnostic uniform aggregation,
120 which grants hallucinated trajectories the same gradient magnitude as flawless ones. Inspired by
121 advantage-weighted regression [Peters and Schaal, 2007, Xue et al., 2026], PRISM resolves this
122 by dynamically prioritizing sampled trajectories via a softmax temperature scaling anchored to
123 programmatic Graph-F1 scores. Finally, because pure token-level matching (e.g., JSD) lacks a
124 sequence-level margin to explicitly penalize truncation [Rashid et al., 2025, Zhou et al., 2024],
125 PRISM hybridizes token-level regularization with sequence-level contrastive preferences [Feng et al.,
126 2025]. By synergizing asymmetric distillation with a reference-free, SimPO-style Length-Normalized
127 Auxiliary Preference Loss [Meng et al., 2024, Wu et al., 2025], the framework optimally balances
128 fine-grained topological routing with global structural completeness.

129 3 Methodology

130 **Problem Formulation.** Let $\mathcal{D} = \{I_i, x_i, y_i^*\}_{i=1}^N$ denote a dataset of ancient-document images
131 I , textual prompts x , and ground-truth OracleGraph JSON structures y^* . MLLMs frame this
132 as autoregressive generation, outputting tokens $y = (y_1, \dots, y_T)$ conditioned on a multimodal
133 context $\mathbf{h}^B = [E_{\text{vis}}(I); E_{\text{txt}}(x)]$. While Supervised Fine-Tuning (SFT) establishes a strong structural
134 prior, it relies on behavioral cloning and lacks a mechanism to rank parse candidates. Standard
135 scalar-reward RL (e.g., DPO, GRPO) suffers from a *short-completion bias*, heavily penalizing long
136 sequences due to zero-reward parsing cliffs. To overcome this, we introduce **PRISM** (illustrated in
137 Figure 1), an asymmetric self-distillation framework that synergizes dense token-level distillation

138 with a sequence-level length-normalized preference margin, governed by a deterministic Graph-F1
 139 reward.

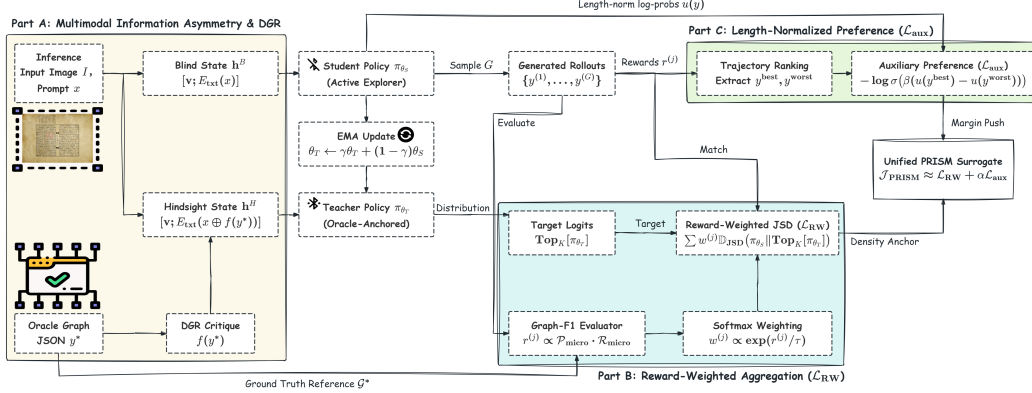


Figure 1: **The Architecture of the PRISM Framework.** (Part A) **Information Asymmetry & DGR:** The *Teacher Policy* evaluates the multimodal context augmented with a Diagnostic Graph Report (DGR) derived from the ground-truth JSON y^* , establishing a strong structural prior without exposing it to the zero-reward cliff. The *Student Policy* relies solely on the blind visual input I to explore and generate G rollouts. (Part B) **Reward-Weighted Aggregation ($\mathcal{L}_{RW}$):** Rollout trajectories are rigorously evaluated via a programmatic *Graph-F1* metric over node and edge topologies. The resulting continuous rewards are transformed via a softmax temperature scaling to dynamically weight the top- K JSD distillation objective, ensuring hallucinated sequences receive minimal gradient priority. (Part C) **Auxiliary Preference Injection ($\mathcal{L}_{aux}$):** To actively push winner-loser margins and explicitly penalize sequence truncation, a SimPO-style length-normalized preference loss is computed over the best and worst trajectories using the active student’s log-probabilities. (Part D) **EMA Dynamics:** To stabilize long-horizon self-distillation, the Teacher’s parameters track the active Student via an Exponential Moving Average ($\gamma=0.999$), providing a dynamically-adjusted yet noise-resilient structural anchor.

140 3.1 Information Asymmetry and the Hindsight Teacher

141 Standard Self-Distillation Policy Optimization (SDPO) assumes information symmetry, trapping the
 142 policy in blind exploration. PRISM introduces a deliberate *information asymmetry* (Figure 1, Part A)
 143 to rescue the Student from zero-reward cliffs without external reward models. During optimization, the
 144 active **Blind Student** π_{θ_S} samples $G = 2$ independent graph rollouts $\{y^{(1)}, \dots, y^{(G)}\}$ conditioned
 145 strictly on $\mathbf{h}^B$.

146 Conversely, the **Hindsight Teacher** π_{θ_T} is granted privileged access to the ground-truth topology via
 147 a **Diagnostic Graph Report (DGR)**—a deterministic natural language critique $f(y^*)$ enumerating
 148 structural constraints and error types. The Teacher’s context becomes $\mathbf{h}^H = [\mathbf{v}; E_{\text{txt}}(x \oplus f(y^*))]$.
 149 Conditioned on $\mathbf{h}^H$, the Teacher is pushed into a high-confidence, low-entropy subspace. To distill
 150 this knowledge, we minimize the token-wise Jensen-Shannon Divergence (JSD). To prevent distilling
 151 long-tail noise, we apply a hard truncation operator $\text{Top}_K[\cdot]$ ($K=20$):

$$\mathcal{L}_{\text{JSD}}^{(j)}(\theta_S) = \sum_{t=1}^{|y^{(j)}|} \mathbb{D}_{\text{JSD}}\left(\pi_{\theta_S}(y_t | \mathbf{h}^B, y_{<t}^{(j)}) \parallel \text{Top}_K[\pi_{\theta_T}(y_t | \mathbf{h}^H, y_{<t}^{(j)})]\right) \quad (1)$$

152 3.2 Deterministic Topological Evaluation via Graph-F1

153 Vanilla SDPO aggregates rollout losses uniformly, treating hallucinated topologies equally with
 154 flawless ones. To resolve this, we map sequences into an optimizable reward space. A parsing
 155 function maps the token sequence to a graph $\mathcal{G}^{(j)} = (\mathcal{V}^{(j)}, \mathcal{E}^{(j)})$. Unparseable sequences receive a
 156 terminal penalty $r^{(j)} = 0$.

For node matching, a predicted node $\hat{v}$ is a True Positive (TP_V) if it shares the exact class with a ground-truth node v^* and their bounding-box Intersection-over-Union (IoU) exceeds $\delta=0.5$ (resolved via one-to-one greedy assignment). An edge $\hat{e}(\hat{u}, \hat{v}, c_{\hat{e}})$ is a True Positive (TP_E) strictly when both its source and target nodes are successfully matched to the corresponding gold nodes, and the relational type matches ($c_{\hat{e}} = c_{e^*}$). The global structural alignment, **Graph Reward** $r^{(j)} \in [0, 1]$, is the micro-averaged F1 score over both node and edge sets:

$$r^{(j)} = \frac{2 \cdot \mathcal{P}_{\text{micro}} \cdot \mathcal{R}_{\text{micro}}}{\mathcal{P}_{\text{micro}} + \mathcal{R}_{\text{micro}} + \epsilon}, \quad \text{where } \mathcal{P}_{\text{micro}} = \frac{|\text{TP}_V| + |\text{TP}_E|}{|\mathcal{V}^{(j)}| + |\mathcal{E}^{(j)}|} \quad (2)$$

and $\mathcal{R}_{\text{micro}}$ is computed symmetrically against the ground-truth objects $|\mathcal{V}^*| + |\mathcal{E}^*|$.

3.3 Reward-Aware Aggregation and Preference Injection

Reward-Weighted Aggregation. As visually detailed in Figure 1 (Part B), we dynamically prioritize sampled trajectories via a softmax temperature scaling ($\tau = 0.5$) anchored to the programmatic Graph-F1 score: $w^{(j)} = \exp(r^{(j)}/\tau) / \sum_{k=1}^G \exp(r^{(k)}/\tau)$. This biases the distillation gradient toward structurally stronger trajectories while avoiding premature collapse on weaker rollouts:

$$\mathcal{L}_{\text{RW}}(\theta_S) = \sum_{j=1}^G w^{(j)} \cdot \mathcal{L}_{\text{JSD}}^{(j)}(\theta_S) \quad (3)$$

Length-Normalized Auxiliary Preference Loss. Pure token-level JSD lacks a sequence-level margin-pushing mechanism, rendering it vulnerable to the short-completion bias. We inject a SimPO-style [Meng et al., 2024] length-normalized auxiliary loss (Figure 1, Part C). Let y^{best} and y^{worst} denote the empirically best and worst rollouts. Using the length-normalized sequence log-probability $u(y) = \frac{1}{|y|} \sum_{t=1}^{|y|} \log \pi_{\theta_S}(y_t | \mathbf{h}^B, y_{<t})$, we formulate:

$$\mathcal{L}_{\text{aux}}(\theta_S) = -\log \sigma\left(\beta(u(y^{\text{best}}) - u(y^{\text{worst}}))\right) \quad (4)$$

where $\beta = 2.5$. The $1/|y|$ factor mathematically counteracts the autoregressive length penalty, explicitly penalizing truncation.

Unified Surrogate Objective. The surrogate objective merges both mechanisms with a preference coefficient $\alpha = 0.2$. To guarantee stable off-policy optimization, we enforce a PPO-style importance clipping $\rho_t(\theta_S) = \frac{\pi_{\theta_S}(y_t | \mathbf{h}^B)}{\pi_{\theta_{\text{old}}}(y_t | \mathbf{h}^B)}$ with boundary $\epsilon = 0.1$. The final objective to be maximized is:

$$\mathcal{J}_{\text{PRISM}}(\theta_S) = \sum_{t=1}^{|y|} \min\left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t\right) - \left(\mathcal{L}_{\text{RW}}(\theta_S) + \alpha \mathcal{L}_{\text{aux}}(\theta_S)\right) \quad (5)$$

where $\hat{A}_t := r^{(j)} - \bar{r}^{(-j)}$ is the leave-one-out sequence-level advantage, applied uniformly to all tokens in rollout j .

3.4 Temporal Stabilization via EMA Dynamics

A strictly frozen Teacher suffers from severe out-of-distribution drift, while an instantaneous online update ($\theta_T \equiv \theta_S$) eradicates the information asymmetry. PRISM solves this target divergence dilemma via an Exponential Moving Average (EMA) update (Figure 1, Part D). The Teacher acts as an exponentially decayed temporal ensemble of all historical Student states:

$$\theta_T \leftarrow \gamma \theta_T + (1 - \gamma) \theta_S \quad (6)$$

With a slow decay rate of $\gamma = 0.999$, the Teacher naturally absorbs and neutralizes transient topological hallucinations generated by the active Student, providing a dynamically-adjusted yet pristine structural prior.

4 Experiments

To empirically validate PRISM and systematically dissect the bottlenecks of standard RL in structured generation, our evaluation is designed across four progressively deeper layers: (1) **Diagnostic Robustness** against zero-shot MLLMs on a 12-dimensional VQA suite; (2) **Matched-rank RL Comparison** to isolate algorithmic gains from parameter capacity; (3) **Learning Dynamics** to reveal the short-completion bias; and (4) **Ablation Studies** under controlled exploratory protocols to attribute the impact of individual architectural components.

4.1 Experimental Setup and Baseline Robustness

Benchmark Composition and Metrics. We introduce **OracleGraph**, a 10,000-page 2.5D historical document benchmark. Unlike standard 2D OCR datasets, OracleGraph utilizes a **Layered Spatial-Semantic Coordinate System**. Each node is augmented with a discrete occlusion-layer index $z \in \{0, \dots, 4\}$, capturing the physical layering of vermilion seals ($z = 4$) over ink text ($z = 1$). The ontology encompasses 13 node classes and 8 directed edge types. The corpus is divided into: (i) 5,000 hand-annotated pages for SFT warmup and RL alignment; (ii) a 500-page held-out VQA audit split; and (iii) a 4,500-page synthetic stress-test split generated via a layout diversity engine. Recognizing that standard linguistic metrics fail to capture strict topological dependencies, we optimize and report **Graph-F1**—a deterministic macro-F1 over node and edge classes.

Zero-Shot Baseline Robustness. To establish the necessity of our fine-tuning approach, we first audit existing MLLMs on a 12-dimensional VQA suite spanning *Basic Logic*, *Global Elements*, and *Advanced 2D Spatial* reasoning (see Table 1). We observe a stark **Structural Specialization** gap: on the 8-metric structural subset (e.g., cross-column jumps), our PRISM-aligned 8B stack reaches 17.1% accuracy, outperforming the strongest same-size open-weight generalist (Qwen2.5-VL-7B, 9.4%) by $1.8\times$. While generalist models excel at *Global Elements* (e.g., EXR long-form extraction at 53.88%) due to their web-scale pretraining, they fail catastrophically on *Advanced Spatial* tasks where our typed-graph prior provides a critical advantage. This confirms that off-the-shelf models lack the structural priors required for complex non-linear layouts, necessitating our RL-aligned pipeline.

Table 1: Diagnostic VQA Evaluation on OracleGraph (Accuracy, % $\uparrow$). **Avg** is the unweighted 12-metric mean; **Win/12** counts metrics strictly outperforming Qwen2.5-VL-7B. PRISM successfully optimizes spatial topology without catastrophic forgetting of general VQA capabilities.

Model	Size	Basic Logic		Global Elements				Advanced 2D Spatial					Avg.	Win/12	
		BVR	HCE	NIR	GQR	EXR	SCR	SVC	CCB	RVR	AGC	MBCV			EFD
Closed-Source API (for reference; thinking-augmented / undisclosed size)															
GPT-4o-mini	-	3.72	1.85	28.38	7.41	21.13	50.00	1.04	2.76	12.24	0.00	3.86	4.07	11.37	2/12
Gemini-2.5-Flash-Thinking [†]	-	40.61	28.50	53.74	22.83	75.90	32.97	7.80	39.94	75.41	13.80	10.73	19.22	35.12	10/12
Open-Weight 8B–12B, Non-Reasoning (apples-to-apples comparison set)															
Llama-3.2-11B-Vision-Ins	11B	0.87	0.00	1.70	5.35	5.55	13.64	1.00	1.80	1.48	0.00	3.51	0.00	2.91	0/12
Gemma-3-12B-IT	12B	2.00	1.03	2.31	8.00	26.53	9.45	1.46	3.52	0.83	0.99	1.70	2.47	5.02	0/12
InternVL3-8B	8B	3.37	4.79	0.00	2.07	6.64	3.90	3.42	3.09	2.22	1.37	6.91	1.54	3.28	0/12
InternVL2.5-8B	8B	3.06	7.59	0.81	2.77	8.39	5.82	1.41	1.62	2.15	2.24	6.48	0.27	3.55	0/12
Qwen2.5-VL-7B-Ins	7B	21.91	4.16	55.05	5.99	53.88	40.45	3.38	14.25	15.97	1.98	5.09	4.70	18.90	ref
Ovis2-8B	8B	8.25	2.10	21.87	4.03	36.97	55.20	6.69	3.10	15.29	1.08	4.71	2.12	13.45	2/12
Open-weight subset mean	–	6.58	3.28	13.62	4.70	22.99	21.41	2.89	4.56	6.32	1.28	4.73	1.85	7.85	–
Open-weight subset max	–	21.91	7.59	55.05	8.00	53.88	55.20	6.69	14.25	15.97	2.24	6.91	4.70	21.03	–
Domain-Specific Fine-Tuned MLLMs (Ours, Open-Weight, Non-Reasoning, 8B)															
Qwen3-VL-8B (Base, no adapter)	8B	38.00	20.00	61.00	7.00	25.94	40.00	3.00	21.00	33.00	5.00	10.00	17.00	23.41	9/12
Qwen3-VL-8B (SFT Baseline)	8B	52.00	20.00	60.00	9.00	27.99	44.00	1.00	19.00	35.00	5.00	17.00	24.00	26.17	10/12
PRISM (seed=42, step 100)	8B	52.00	20.00	60.00	9.00	27.63	44.00	1.00	20.00	34.00	3.00	17.00	25.00	26.05	10/12
PRISM (seed=2024, step 180)	8B	51.00	20.00	60.00	8.00	27.64	44.00	1.00	20.00	35.00	3.00	17.00	24.00	25.89	10/12

[†] Gemini-2.5-Flash-Thinking: closed API, uses inference-time chain-of-thought reasoning, total parameter count undisclosed. Listed for context, excluded from our open-weight 8B–12B non-reasoning apples-to-apples comparison. On the EXR long-form extraction metric alone its 75.90 leaves a large gap to all 8B open-weight models, consistent with its thinking / larger-size advantage.

4.2 Main RL Results: The Matched-Rank Regime

To rigorously assess RL alignment without the confounding effects of parameter scaling or unequal compute, we evaluate PRISM and existing RL baselines under a strict matched LoRA rank ($r=32$) and matched step (100 steps) protocol. Results are summarized in Table 2, revealing three distinct behavioral regimes.

Regime I: The SFT Attractor (DPO, GRPO, SimPO). Standard preference-based methods cluster within ± 1.5 pp of the SFT baseline (72.94%) without achieving statistical significance. We interpret

this as a *policy-level signal gap*: scalar rewards collapse complex structural graph errors into a single dimension, trapping the policy near the SFT attractor. Furthermore, these methods actively shrink their response lengths (down to ~ 1312 tokens) to minimize parsing-error risks, demonstrating a severe **Short-Completion Bias**.

Regime II: Schema Collapse (Vanilla-SDPO). Without PRISM’s structural critique and preference mechanisms, Vanilla-SDPO regresses by -2.79pp . Its schema-parsed rate drops to 94.0%, proving that relying on a raw EMA teacher without diagnostic guidance forces the student into structural instability.

Regime III: Robust Optimization (PRISM). PRISM uniquely breaks the plateau, exceeding the SFT mean (73.96%) while simultaneously achieving the lowest cross-seed standard deviation (0.27pp, a $2.4\times$ reduction over SFT). It perfectly balances long-completion stability (1431 tokens) with topological accuracy. Specifically, PRISM achieves a highly significant $+3.81\text{pp}$ gain over Vanilla-SDPO ($p < 0.001$), successfully isolating the necessity of our reward-weighted aggregation and auxiliary preference mechanisms.

Table 2: RL comparison under a matched-rank ($r=32$) and matched-compute (100 steps) protocol. PRISM uniquely exceeds the SFT mean while significantly reducing cross-seed optimization variance.

Method (matched $r=32$, 100 steps)	Seeds	$n=100$ held-out OracleGraph eval					
		Reward	Cross-seed std	Δ vs. SFT	p vs SFT	Parsed %	Resp. Len
Base (Qwen3-VL-8B, no adapter)	1	13.00%	—	-59.94pp	—	—	—
SFT baseline ($r=32$, 1000 steps)	3	72.94%	0.65pp	0	—	98.0%	1337
DPO (matched $r=32$)	3	73.64%	0.39pp	$+0.70\text{pp}$	0.33	98.3%	1312
GRPO (matched $r=32$)	3	73.74%	1.77pp	$+0.80\text{pp}$	0.41	98.0%	1338
SimPO (matched $r=32$)	3	72.65%	1.05pp	-0.29pp	0.68	96.7%	1390
Vanilla-SDPO ($r=32$, no DGR, no RW, no aux pref.)	3	70.15%	1.73pp	-2.79pp	<0.01 (worse)	94.0%	1555
Naive-hp SDPO ($r=32$, text-LLM hyperparameters)	1	63.3%	—	-9.64pp	—	90.0%	1787
DGR-long (single seed)	1	73.33%	—	$+0.39\text{pp}$	—	97.0%	1447
PRISM (ours, full: SDPO + DGR + RW + aux pref, $r=32$)	3	73.96%	0.27pp	$+1.02\text{pp}$	0.12	96.3%	1431
Controlled intra-family ablation:		PRISM vs. Vanilla-SDPO $\Delta = +3.81\text{pp}$ Graph Reward, 95% CI $[+1.95, +5.76]\text{pp}$, $p < 0.001$.					

Evaluated via 10,000-resample paired bootstrap. PRISM achieves a highly significant $+3.81\text{pp}$ gain over Vanilla-SDPO, isolating the necessity of the proposed reward-weighted aggregation and auxiliary preference mechanisms.

4.3 Learning Dynamics and DGR Verbosity Analysis

To trace the origins of PRISM’s stability and the baselines’ failures, we examine the continuous training dynamics and explicit Teacher critiques.

Learning Dynamics and Optimization Stability. Visualizing the continuous RL optimization horizon in Figure 2, we definitively capture the behavioral divergence between PRISM and standard scalar-reward RL. First, the Top- K JSD training loss (Fig. 2a) drops monotonically and stabilizes smoothly, confirming that the reward-weighted distillation objective safely aligns the Student to the Teacher despite the strict information asymmetry. Second, examining the rollout-reward trajectory (Fig. 2b) reveals that standard algorithms (DPO, GRPO) rapidly plateau with high oscillatory variance, struggling to close the performance gap. This oscillation is a downstream symptom of the **Short-Completion Bias**: baselines shrink their outputs to evade zero-reward parser failures, leading to unstable reward signals. Conversely, PRISM climbs fastest and converges highest, effectively avoiding catastrophic truncation and staying on the Pareto frontier.

DGR Verbosity and Teacher Critique. To evaluate whether the Teacher’s structured critique is an independent lever, we sweep the Diagnostic Graph Report (DGR) verbosity under the matched- $r=32$ protocol (Table 3). The entire DGR family empirically dominates non-DGR baselines. The auto configuration, which dynamically routes verbosity based on the student’s error type, achieves the lowest cross-seed standard deviation (0.27pp), confirming that providing targeted structural statistics prevents text-overfitting noise.

4.4 Mechanistic Analysis and Ablations (Stage-5 Protocol)

To systematically dissect the internal drivers of PRISM’s success and rule out external confounders, we conduct four targeted mechanistic analyses. **Protocol Note:** These qualitative ablations are evaluated under a controlled *Stage-5 exploratory protocol* ($n=30$ pages, unmatched LoRA rank,

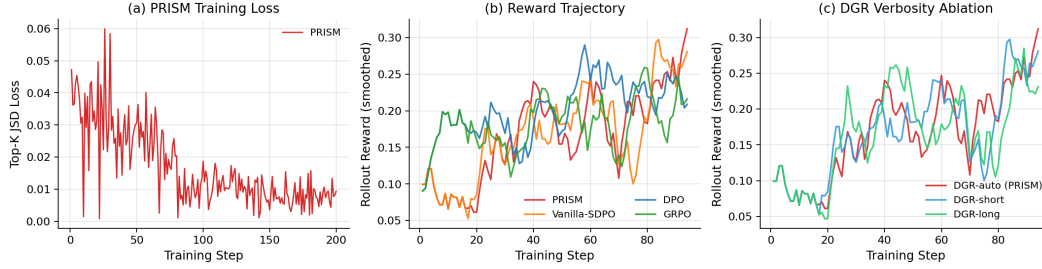


Figure 2: **Learning Dynamics of the PRISM recipe.** (a) Top- K JSD training loss of the PRISM main run (seed 42, 200 steps). The loss decreases monotonically and flattens after step 150, confirming the reward-weighted teacher-distillation objective is well-behaved despite asymmetric DGR conditioning. (b) Smoothed rollout-reward trajectory (centred moving average, $k=7$) across four RL families on the first 100 steps of training. PRISM climbs fastest and converges highest; SDPO (visualised here by the short-DGR proxy run from the ablation suite; per-seed validation numbers in Table 2 are authoritative) plateaus below; DPO / GRPO oscillate without closing the gap. (c) DGR-verbosity ablation on the rollout-reward trajectory: DGR-auto (PRISM’s default length-adaptive report) dominates both fixed short and fixed long variants, motivating auto as the production configuration.

Table 3: DGR verbosity ablation (matched- $r=32$). The auto configuration dynamically adjusts feedback based on the error type, yielding the most stable optimization across seeds.

DGR variant	Feedback axes						Graph Reward (3-seed)	Seeds
	P/F	Macro	PerType	Diffs	Gold	Examples		
None (Vanilla SDPO, DGR off)	×	×	×	×	×	×	70.15% $\pm$ 1.41pp	3
L0 (pass/fail only)	✓	×	×	×	×	×	73.49% $\pm$ 1.04pp	3
L2 (+ macro/per-type)	✓	✓	✓	×	×	×	74.59% $\pm$ 1.14pp	3
L3 (+ diff items)	✓	✓	✓	✓	×	×	73.77% $\pm$ 0.67pp	3
short (gold + diff)	×	×	✓	✓	✓	×	73.09% $\pm$ 0.93pp	3
long (gold + diff + ex.)	×	×	✓	✓	✓	✓	73.12% $\pm$ 0.21pp	3
auto (PRISM, Ours)	Dynamic	Dyn.	✓	✓	✓	Dyn.	73.96% $\pm$ 0.27pp	3

512-token bottleneck) to rapidly isolate architectural behaviors. Therefore, absolute rewards are scaled differently and are not directly comparable to the highly optimized results in §4.2.

(1) **Teacher Asymmetry (Table 4).** PRISM fundamentally relies on the information asymmetry between a gold-conditioned teacher and a gold-blind student. Ablating the update rule shows that an online control (where the student effectively teaches itself) collapses below the untrained base model (0.061). The EMA strategy uniquely tracks the student’s distribution while preserving a robust structural anchor.

Table 4: Teacher-Update Mechanisms. The online nullity control confirms that self-distillation fails without strict information asymmetry. (Untrained base model scores 0.130).

Teacher Mechanism	Graph Reward $\uparrow$	Δ vs. Base
online (<i>teacher</i> $\equiv$ <i>student</i>)	0.061	-0.069
frozen (Fixed at step 0)	0.228	+0.098
trust-region	0.265	+0.135
EMA (Ours, $\alpha = 0.995$)	0.429	+0.299

(2) **Topological Recovery (Table 5).** To identify *where* scalar RL fails, we decompose the Graph-F1 into specific node and edge classes. While simple text blocks (CF) are robust across all models, standard DPO and GRPO catastrophically fail on dense relational topologies: Main Text (MT) bounding boxes and Reads-After (R) sequential edges. PRISM (SDPO) is the only framework that effectively recovers these complex spatial categories, trailing only the fully-supervised SFT reference.

(3) **Optimization Stability (Table 6).** Optimizing massive JSON sequences is highly sensitive to gradient noise. Optimal performance is strictly bounded to a rollout group size of $g = 2$ and an

Table 5: Fine-Grained Per-Class F1 Analysis. PRISM recovers complex spatial dependencies (Main Text MT, Reads-After R) where standard scalar RL (DPO/GRPO) completely collapses.

Method	Node Types (F1)					Edge Types (F1)		
	MT	IA	TM	CF	RS	R	A	O
Untrained Base	0%	10%	40%	85%	24%	0%	0%	0%
DPO (Baseline)	0%	0%	0%	0%	0%	0%	0%	0%
GRPO (Baseline)	0%	3%	36%	100%	13%	0%	0%	0%
SDPO (Ours)	72%	12%	75%	100%	33%	33%	2%	3%
<i>SFT (Upper Ref.)</i>	84%	74%	81%	100%	93%	52%	10%	22%

importance clip of $\epsilon = 0.2$. Removing within-group contrast ($g = 1$) or utilizing unconstrained variance ($\epsilon = 0$) severely degrades the reward, validating our mathematically constrained off-policy formulation.

Table 6: RL Optimization Stability Ablation. Both within-group advantages ($g=2$) and strict importance clipping ($\epsilon=0.2$) are critical to preventing gradient divergence.

Hyperparameter	Configuration Variant	Graph Reward $\uparrow$
Group Size (g)	$g = 1$ (No in-group contrast)	0.256
	$g = 3$ (High optimization noise)	0.306
	$g = 2$ (Ours, best in scan)	0.429
Importance Clip (ϵ)	$\epsilon = 0$ (Unconstrained variance)	0.249
	$\epsilon = 0.2$ (Ours, stabilized)	0.429

(4) Capacity vs. Algorithm (Table 7). To conclusively prove that our gains stem from reward-aware signals rather than mere parameter scaling, we ablated the LoRA rank (r) on a merged SFT base. Naively increasing model capacity ($r = 32, 64$) directly leads to catastrophic representation collapse (Graph Reward dropping to 14.9%). This isolates PRISM’s efficacy to its dense gradient formulation rather than parameter inflation.

Table 7: Algorithm vs. Parameter Capacity Attribution. Naively increasing the LoRA rank ($r = 32, 64$) on a merged SFT base model fails catastrophically.

Configuration (merged base, Stage-5)	Graph Reward $\uparrow$	CER $\downarrow$	BLEU $\uparrow$
PRISM ($r = 16$ on merged base)	71.8%	0.084	0.892
SDPO ($r = 32$ on merged base)	19.4%	0.353	0.643
SDPO ($r = 64$ on merged base)	14.9%	0.319	0.663

5 Conclusion, Limitations, and Future Work

We introduce **OracleGraph**, a 10,000-page 2.5D benchmark equipped with a fine-grained hallucination taxonomy and a 12-dimensional VQA suite for ancient Chinese documents. Our diagnosis reveals standard scalar-reward RL (e.g., DPO, GRPO) struggles to surpass SFT baselines due to a *policy-level signal gap* and *short-completion bias* in long JSON generation.

We propose **PRISM**, an asymmetric self-distillation framework. Synergizing a schema-aware Hindsight Teacher with reward-weighted aggregation and a length-normalized preference loss, PRISM suppresses catastrophic truncation. It yields a +3.81pp Graph-F1 gain over vanilla distillation ($p < 0.001$) and uniquely stabilizes cross-seed variance.

Limitations. First, our OOD evaluation couples layout and source shifts; pure cross-book generalization remains open. Second, PRISM’s +1.02pp gain over SFT ($p=0.12$) highlights the SFT ceiling on complex topologies. Third, uniform reward-weighting amplifies noise on pathologically hard pages, motivating adaptive confidence gating. Finally, extending from historical Chinese to multilingual manuscripts is a promising digital humanities direction.

References

- Mohammad Beigi, Ming Jin, Junshan Zhang, Qifan Wang, and Lifu Huang. Adversarial reward auditing for active detection and mitigation of reward hacking. *CoRR*, abs/2602.01750, 2026. doi: 10.48550/ARXIV.2602.01750. URL <https://doi.org/10.48550/arXiv.2602.01750>.
- Franz Louis Cesista. Multimodal structured generation: Cvpr’s 2nd mmfm challenge technical report, 2025. URL <https://arxiv.org/abs/2406.11403>.
- Lei Chen, Xuanle Zhao, Zhixiong Zeng, Jing Huang, Liming Zheng, Yufeng Zhong, and Lin Ma. Breaking the sft plateau: Multimodal structured reinforcement learning for chart-to-code generation, 2026a. URL <https://arxiv.org/abs/2508.13587>.
- Wei Chen, Yubing Wu, Junmei Yang, Delu Zeng, Qibin Zhao, John Paisley, Min Chen, and Zhou Wang. Towards disentangled preference optimization dynamics beyond likelihood displacement, 2026b. URL <https://arxiv.org/abs/2604.18239>.
- Yufan Chen, Ruiping Liu, Junwei Zheng, Di Wen, Kunyu Peng, Jiaming Zhang, and Rainer Stiefelha-gen. Graph-based document structure analysis. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=Fu0aggezN9>.
- Ruiyi Ding, Yongxuan Lv, Xianhui Meng, Jiahe Song, Chao Wang, Chen Jiang, and Yuan Cheng. Prpo: Aligning process reward with outcome reward in policy optimization, 2026. URL <https://arxiv.org/abs/2601.07182>.
- Wu Fei, Hao Kong, Shuxian Liang, Yang Lin, Yibo Yang, Jing Tang, Lei Chen, and Xiansheng Hua. Self-guided process reward optimization with redefined step-wise advantage for process reinforcement learning, 2025. URL <https://arxiv.org/abs/2507.01551>.
- Zhili Feng, Dhananjay Ram, Cole Hawkins, Aditya Rawal, Jinman Zhao, and Sheng Zha. Sequence-level large language model training with contrastive preference optimization, 2025. URL <https://arxiv.org/abs/2502.16433>.
- Yanggan Gu, Junzhuo Li, Sirui Huang, Xin Zou, Zhenghua Li, and Xuming Hu. Capturing nuanced preferences: Preference-aligned distillation for small language models, 2025. URL <https://arxiv.org/abs/2502.14272>.
- Anwen Hu, Haiyang Xu, Jiabo Ye, Ming Yan, Liang Zhang, Bo Zhang, Chen Li, Ji Zhang, Qin Jin, Fei Huang, and Jingren Zhou. mplug-docowl 1.5: Unified structure learning for ocr-free document understanding, 2024. URL <https://arxiv.org/abs/2403.12895>.
- Jonas Hübner, Frederike Lübeck, Lejs Behric, Anton Baumann, Marco Bagatella, Daniel Marta, Ido Hakimi, Idan Shenfeld, Thomas Kleine Buening, Carlos Guestrin, and Andreas Krause. Reinforcement learning via self-distillation, 2026. URL <https://arxiv.org/abs/2601.20802>.
- Qi Ju, Falin Hei, Zhemei Fang, and Yunfeng Luo. Elo-rated sequence rewards: Advancing reinforcement learning models, 2025. URL <https://arxiv.org/abs/2409.03301>.
- Chen Li, Han Zhang, Zhantao Yang, Fangyi Chen, Zihan Wang, Anudeepsekhar Bolimera, and Marios Savvides. Stellar-vision: Self-topology-aware efficient learning for aligned reasoning in vision, 2026a. URL <https://arxiv.org/abs/2508.08688>.
- Gengsheng Li, Tianyu Yang, Junfeng Fang, Mingyang Song, Mao Zheng, Haiyun Guo, Dan Zhang, Jinqiao Wang, and Tat-Seng Chua. Unifying group-relative and self-distillation policy optimization via sample routing, 2026b. URL <https://arxiv.org/abs/2604.02288>.
- Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan-gao Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe, 2026c. URL <https://arxiv.org/abs/2604.13016>.
- Peng Liao, Peijia Zheng, Lingbo Li, Shangsong Liang, and Lin Chen. Intrinsic mutual information as a modulator for preference optimization, 2026. URL <https://arxiv.org/abs/2604.24804>.

- 343 Chengqi Lyu, Songyang Gao, Yuzhe Gu, Wenwei Zhang, Jianfei Gao, Kuikun Liu, Ziyi Wang,
344 Shuaibin Li, Qian Zhao, Haian Huang, Weihao Cao, Jiangning Liu, Hongwei Liu, Junnan Liu,
345 Songyang Zhang, Dahua Lin, and Kai Chen. Exploring the limit of outcome reward for learning
346 mathematical reasoning, 2025. URL <https://arxiv.org/abs/2502.06781>.
- 347 Monte MacDiarmid, Benjamin Wright, Jonathan Uesato, Joe Benton, Jon Kutasov, Sara Price, Naia
348 Bouscal, Sam Bowman, Trenton Bricken, Alex Cloud, Carson Denison, Johannes Gasteiger, Ryan
349 Greenblatt, Jan Leike, Jack Lindsey, Vlad Mikulik, Ethan Perez, Alex Rodrigues, Drake Thomas,
350 Albert Webson, Daniel Ziegler, and Evan Hubinger. Natural emergent misalignment from reward
351 hacking in production rl, 2025. URL <https://arxiv.org/abs/2511.18397>.
- 352 Yu Meng, Mengzhou Xia, and Danqi Chen. Simpo: Simple preference optimization with a reference-
353 free reward, 2024. URL <https://arxiv.org/abs/2405.14734>.
- 354 Jan Peters and Stefan Schaal. Using reward-weighted regression for reinforcement learning of task
355 space control. *2007 IEEE International Symposium on Approximate Dynamic Programming and*
356 *Reinforcement Learning*, pages 262–267, 2007. URL [https://api.semanticscholar.org/](https://api.semanticscholar.org/CorpusID:14180705)
357 [CorpusID:14180705](https://api.semanticscholar.org/CorpusID:14180705).
- 358 Deng Qiyuan, Kehai Chen, Min Zhang, and Zhongwen Xu. HiPO: Self-hint policy optimization for
359 RLVR. In *The Fourteenth International Conference on Learning Representations*, 2026. URL
360 <https://openreview.net/forum?id=rcb20pHmT1>.
- 361 Ahmad Rashid, Ruotian Wu, Julia Grosse, Agustinus Kristiadi, and Pascal Poupart. A critical look at
362 tokenwise reward-guided text generation, 2025. URL <https://arxiv.org/abs/2406.07780>.
- 363 Massimo Rizzoli, Simone Alghisi, Olha Khomyn, Gabriel Roccabruna, Seyed Mahed Mousavi,
364 and Giuseppe Riccardi. Civet: Systematic evaluation of understanding in vlms, 2025. URL
365 <https://arxiv.org/abs/2506.05146>.
- 366 Tim G. J. Rudner, Vitchyr H. Pong, Rowan Thomas McAllister, Yarín Gal, and Sergey Levine.
367 Outcome-driven reinforcement learning via variational inference. In A. Beygelzimer, Y. Dauphin,
368 P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*,
369 2021. URL <https://openreview.net/forum?id=4bzanicqvy8>.
- 370 Haozhe Wang, Chao Qu, Zuming Huang, Wei Chu, Fangzhen Lin, and Wenhui Chen. VI-rethinker:
371 Incentivizing self-reflection of vision-language models with reinforcement learning, 2025. URL
372 <https://arxiv.org/abs/2504.08837>.
- 373 Xiaohua Wang, Muzhao Tian, Yuqi Zeng, Zisu Huang, Jiakang Yuan, Bowen Chen, Jingwen Xu,
374 Mingbo Zhou, Wenhao Liu, Muling Wu, Zhengkang Guo, Qi Qian, Yifei Wang, Feiran Zhang,
375 Ruicheng Yin, Shihan Dou, Changze Lv, Tao Chen, Kaitao Song, Xu Tan, Tao Gui, Xiaoqing
376 Zheng, and Xuanjing Huang. Reward hacking in the era of large models: Mechanisms, emergent
377 misalignment, challenges, 2026. URL <https://arxiv.org/abs/2604.13602>.
- 378 Han Weng, Puzhen Wu, Cui Longjie, Yi Zhan, Boyi Liu, Yuanfeng Song, Dun Zeng, Yingxiang
379 Yang, Qianru Zhang, Dong Huang, Xiaoming Yin, Yang Sun, and Xing Chen. Graph-reward-SQL:
380 Execution-free reinforcement learning for text-to-SQL via graph matching and stepwise reward.
381 In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors,
382 *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 12917–12943,
383 Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-
384 335-7. doi: 10.18653/v1/2025.findings-emnlp.694. URL [https://aclanthology.org/2025.](https://aclanthology.org/2025.findings-emnlp.694/)
385 [findings-emnlp.694/](https://aclanthology.org/2025.findings-emnlp.694/).
- 386 Junkang Wu, Xue Wang, Zhengyi Yang, Jiancan Wu, Jinyang Gao, Bolin Ding, Xiang Wang, and
387 Xiangnan He. Alphaspo: Adaptive reward margin for direct preference optimization, 2025. URL
388 <https://arxiv.org/abs/2410.10148>.
- 389 Shuchen Xue, Chongjian GE, Shilong Zhang, Yichen Li, and Zhi-Ming Ma. Advantage weighted
390 matching: Aligning RL with pretraining in diffusion models, 2026. URL [https://openreview](https://openreview.net/forum?id=wekWtzTbUP)
391 [.net/forum?id=wekWtzTbUP](https://openreview.net/forum?id=wekWtzTbUP).

- 392 Ning Yang, Hai Lin, Yibo Liu, Baoliang Tian, Guoqing Liu, and Haijun Zhang. Token-importance
393 guided direct preference optimization. In *The Fourteenth International Conference on Learning*
394 *Representations*, 2026. URL <https://openreview.net/forum?id=cMEvMvMw9>.
- 395 Qi Zhang, Shouqing Yang, Lirong Gao, Hao Chen, Xiaomeng Hu, Jinglei Chen, Jiexiang Wang, Sheng
396 Guo, Bo Zheng, Haobo Wang, and Junbo Zhao. Lets: Learning to think-and-search via process-
397 and-outcome reward hybridization, 2025a. URL <https://arxiv.org/abs/2505.17447>.
- 398 Yizhuo Zhang, Heng Wang, Shangbin Feng, Zhaoxuan Tan, Xinyun Liu, and Yulia Tsvetkov. Gener-
399 alizable LLM learning of graph synthetic data with reinforcement learning. *CoRR*, abs/2506.00845,
400 2025b. doi: 10.48550/ARXIV.2506.00845. URL [https://doi.org/10.48550/arXiv.2506.](https://doi.org/10.48550/arXiv.2506.00845)
401 00845.
- 402 Guanghao Zhou, Panjia Qiu, Cen Chen, Jie Wang, Zheming Yang, Jian Xu, and Minghui Qiu.
403 Reinforced mllm: A survey on rl-based reasoning in multimodal large language models, 2025.
404 URL <https://arxiv.org/abs/2504.21277>.
- 405 Wenxuan Zhou, Shujian Zhang, Lingxiao Zhao, and Tao Meng. T-reg: Preference optimization with
406 token-level reward regularization, 2024. URL <https://arxiv.org/abs/2412.02685>.
- 407 Deyu Zou, Yongqiang Chen, Jianxiang Wang, Garry YANG, Mufei Li, Qing Da, James Cheng, Pan
408 Li, and Yu Gong. Reducing belief deviation in reinforcement learning for active reasoning of LLM
409 agents. In *The Fourteenth International Conference on Learning Representations*, 2026. URL
410 <https://openreview.net/forum?id=r8hzDA3pUY>.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: We explicitly demarcate what our method (PRISM) achieves (a +3.81pp intra-family gain and $3\times-5\times$ lower cross-seed std than scalar RL baselines) from what it does not claim (we do not claim PRISM acts as a ceiling-breaking RL method over SFT in-distribution at $p < 0.05$). The empirical observation regarding the *policy-level signal gap* for standard scalar RL (DPO, GRPO, SimPO clustering within ± 1.5 pp of SFT) is stated objectively without over-claiming a single causal mechanism. All claims (C1-C5) in the Introduction are backed by precise $n = 100$, 3-seed paired bootstrap bounds in Section 4.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: We dedicate a specific sub-section to explicit limitations in Section 5 (Limitations and Future Work). We honestly document four constraints: (L1) the preliminary nature of our cross-book OOD evaluation (which is a combined book $\times$ density shift, not a pure cross-book external-validity test); (L2) the lack of statistical significance at the conventional $p < 0.05$ threshold for PRISM’s gain over SFT in-distribution; (L3) the potential for reward-weighting to hurt performance on extremely hard, reward-sparse pages; and (L4) the limits of our 5,000-page dataset in terms of cross-language transfer.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.

- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The contributions of this paper are primarily empirical and data-centric (the OracleGraph benchmark and the empirical diagnosis of RL regimes in structured generation). While we formalize the PRISM surrogate objective and EMA dynamics mathematically in Section 3, these are algorithm designs and empirical heuristics rather than formal theoretical theorems requiring mathematical proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: We release the full 10,000-page OracleGraph dataset, the 12-dimensional VQA diagnostic suite, the deterministic Graph-F1 evaluation script, and the training code for PRISM via an anonymized GitHub repository (<https://anonymous.4open.science/r/PRISM2026>). We explicitly document all critical hyperparameters (e.g., LoRA $r = 32$, EMA $\gamma = 0.999$, importance clip $\epsilon = 0.1$) in Section 4.2 and Appendix 4.4. To ensure rigorous reproducibility, our RL comparison protocol is executed across 3 random seeds under identical compute budgets, evaluated via 10,000-resample paired bootstrap.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We provide an anonymized GitHub repository (<https://anonymous.4open.science/r/PRISM2026>) containing the source code for PRISM, the deterministic Graph-F1 evaluation script, and the task-adaptive VQA scoring framework. We also fully release the **OracleGraph** dataset, including the 5,000-page hand-annotated corpus and the 12-dimensional VQA suite, under a CC-BY-NC-4.0 license. Detailed instructions to reproduce the SFT and RL environments are provided in the README.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The core experimental setting, including the data splits (in-distribution vs. cross-book OOD), the matched-compute protocol, and critical hyperparameters (LoRA $r = 32$, EMA $\gamma = 0.999$, importance clip $\epsilon = 0.1$, $g = 2$, and $\text{max_new_tokens} = 3072$)

are explicitly detailed in Section 3 and the preamble to Section 4.2. The task-adaptive VQA evaluation metric is thoroughly defined in Appendix A.3.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report the cross-seed standard deviation across 3 random seeds for all matched-rank RL comparisons in Table 2 and the OOD probe in Table 15. Furthermore, we establish the statistical significance of our primary intra-family ablation (PRISM vs. Vanilla-SDPO) and other RL baselines using a rigorous **10,000-resample paired bootstrap** protocol on the $n = 100$ held-out set, explicitly reporting both the 95% Confidence Intervals (CIs) and the exact p -values. We also perform a pre-registered sub-group analysis with Holm-Bonferroni family-wise error correction (Appendix B.1).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: We detail the exact hardware requirements in Table 14, noting that our 8B stack is locally deployable for single-batch inference on a single 24GB GPU, or a single H20 (96 GB) / L40 (48 GB) for full training. All RL experiments were conducted on an 8-GPU H20 node. We explicitly state the training budget (e.g., 100 optimization steps) for our matched-compute comparisons in Section 4.2 to ensure fair reproducibility.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research involves the digitization and structural parsing of public-domain ancient Chinese documents (e.g., Ming and Qing woodblock editions). It does not involve human subjects, living individuals' personally identifiable information (PII), or technologies that could inflict societal harm, fully conforming to the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [No]

Justification: We explicitly note the positive impact of our work on the digital humanities community in Section 1 (providing reliable open-weight parsers for ancient corpora). We do not include a dedicated discussion on negative societal impacts because the structural parsing of ancient, public-domain historical manuscripts does not present foreseeable malicious applications (e.g., no privacy, surveillance, or deepfake risks).

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: Our released assets (the OracleGraph dataset and the PRISM layout-parsing models) focus strictly on the structural layout analysis of historical literature. They do not fall under the category of high-risk generative models (such as deepfake generators or open-ended LLMs) or privacy-sensitive datasets that would require specific safety filters or gated release safeguards.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: As explicitly detailed in Section 4.1 (Annotation quality and governance), all source image scans are public-domain Ming and Qing woodblock editions or historical manuscripts for which no contemporary living-author rights apply. We explicitly declare that our resulting OracleGraph benchmark dataset is released under the CC-BY-NC-4.0 license for research use.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: We introduce the **OracleGraph** benchmark and the **PRISM** framework. The dataset composition, typed-graph ontology, and annotation governance are extensively documented in Appendix A.1. Both the assets and their documentation are released via the anonymized GitHub repository.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The dataset was constructed and cross-validated by trained domain experts. It does not involve public crowdsourcing platforms (e.g., MTurk) or human subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: This research focuses entirely on the structural parsing of public-domain ancient Chinese literature (Ming and Qing woodblock editions) and does not involve human subjects. Thus, IRB approval is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

780 Question: Does the paper describe the usage of LLMs if it is an important, original, or
781 non-standard component of the core methods in this research? Note that if the LLM is used
782 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
783 scientific rigor, or originality of the research, declaration is not required.

784 Answer: [N/A]

785 Justification: The core method development in this research does not involve LLMs as any
786 important, original, or non-standard components.

787 Guidelines:

- 788 • The answer [N/A] means that the core method development in this research does not
789 involve LLMs as any important, original, or non-standard components.
- 790 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
791 be described.

A Technical Appendices and Supplementary Material

A.1 The OracleGraph Corpus: Design and Annotation

Our OracleGraph corpus consists of 10,000 high-resolution page images of Chinese ancient documents (principally block-printed novels, historical ledgers, and hand-copied manuscripts), each paired with a hand-verified JSON oracle graph. Beyond raw layout, the annotation specifically captures eight classes of structural ambiguity that we find to be the dominant failure modes of generalist MLLMs on ancient Chinese content:

- **Type I: Layout Hierarchy & Reading Order.** *Double-line annotation mixing* (small-font paired commentary interleaving main text and being hierarchically conflated), *marginalia misalignment* (wrong vertical anchoring of top/bottom/side notes), *cross-column boundary breakage* (mispredicting which column a sentence continues into), *implicit grid collapse* (columns defined only by absence of a rule, not by an explicit line).
- **Type II: Editing Instructions & Symbols.** *Alteration ignorance* (failing to process historical deletions or insertions encoded by edit marks), *inversion-mark ignorance* (ignoring traditional transposition marks), *punctuation misinterpretation* (reading vermilion grammatical dots as lexical tokens).
- **Type III: Multi-modal Interference & Overlay.** *Dense text-image mismatch*, *sub-image relation loss*, *seal intrusion* (red stamps partially occluding black text), and *multi-color hierarchy confusion* (mixing vermilion commentary with standard main text).

Listing 1: Truncated JSON representation of an OracleGraph page. The schema defines bounding-box polygon, semantic node type, and per-node occlusion-layer index.

```
{
  "image_id": "honglouloumeng_page_025",
  "adversarial_scenario": "NOTE_MISANCHORING",
  "metadata": { "width": 1885, "height": 1333 },
  "oracle_graph": {
    "nodes": [
      {
        "node_id": "n_frame1",
        "polygon": [167, 413, 1641, 413, 1641, 1194, 167, 1194],
        "type": "OUTER_PAGE_FRAME",
        "attributes": { "z_index": 0, "color": "black" }
      },
      {
        "node_id": "n_separator1",
        "polygon": [1641, 413, 1641, 1194, 1640, 1194, 1640, 413],
        "type": "COLUMN_SEPARATOR",
        "attributes": { "z_index": 0, "color": "black" }
      }
    ]
    // ... (Text nodes, Seals, and Edges omitted for brevity)
  }
}
```

A.2 Examples of Annotated Pages

The following figures illustrate representative pages from our corpus, each paired in our release with its hand-verified JSON oracle graph:

A.3 A Task-Adaptive Evaluation Framework for Multimodal Large Language Models in Complex Layout Understanding

Traditional exact match paradigms struggle to objectively reflect the true representational capacity of Multimodal Large Language Models (MLLMs) due to their generative heterogeneity—such as redundant responses, inconsistent numerical expressions, and long-text dislocations—when processing visual question answering tasks on high-resolution ancient document layouts. Inspired by recent studies on alignment and model robustness, this research proposes a task-adaptive quantitative evaluation framework (see Table 8). Tailored to the distinct characteristics of various task dimensions,



Figure 3: Page from a block-printed *Water Margin* edition. This layout exhibits Type I (double-line interlinear annotations) and Type III (seal intrusion) structural ambiguities. The paired JSON oracle graph (not shown) encodes the occlusion-layer index and reading-order edges for the overlapping elements.



Figure 4: Page from a block-printed *The Story of the Stone (Hongloumeng)* edition. It features main text with cross-column side marginalia, second-layer marginal comments, and cinnabar seals interspersed with the ink text.

this framework constructs three customized evaluation models to systematically mitigate evaluation bias through the specific mechanisms detailed below.

Model I: Spatial Localization-Oriented Evaluation (Discrete, 0/1). This model is applied to tasks requiring precise coordinate-to-text anchoring, including **BVR, MBCV, SVC, CCB, RVR, and AGC**. Given the tendency of MLLMs to generate verbose explanations, we implement a *Multi-level Fault Tolerance* mechanism:

- **Exact Match:** Standardized comparison after stripping punctuation and whitespace.
- **Constrained Forward Inclusion:** We perform a reverse search for the Ground Truth (GT) anchor within the prediction. To prevent "cheating" via exhaustive generation, the match is only valid if the total response length is within $5\times$ the length of the GT.
- **Reverse Prefix Matching:** If the model's output is a valid prefix substring of the GT, it is counted as a successful physical localization, accounting for premature sequence truncation.

Model II: Semantic Equivalence-Based Numerical Evaluation (Discrete, 0/1). This model targets quantitative tasks such as **GQR (Grid Query)** and **SCR (Seal Counting)**. It resolves "modality aliasing" through a three-step pipeline:

1. **Heuristic Filtering:** Removal of colloquial prefixes to eliminate false-positive interference from characters like "one".
2. **Unified Numerical Mapping:** A positional weight-based decoding graph converts Chinese numeral strings (0–999) into standardized Arabic formats.
3. **Set Verification:** We extract all continuous numerical values using regular expressions. A score of 1.0 is granted if the GT exists within the predicted numerical set.

Model III: Order-Agnostic Continuous Evaluation (Continuous, 0–1). For long-form extraction tasks like **EXR (Full Extraction)**, where OCR perturbations and layout dislocations are common, we move beyond binary judgments to a *character-set F1* metric (P/R/F1 computed over the multiset of characters treated as unordered, not a position-aware token-F1). We make this design choice *explicit* because character-F1 is ambiguous in the document-parsing literature: "character-F1" can refer to position-aware alignment (e.g., Levenshtein-based), multiset-overlap, or unordered-set-overlap, and the three are not equivalent. We use the **unordered-multiset variant** on EXR specifically so that line-swapping or sequence dislocations caused by complex non-linear ancient layouts do not penalize a correct extraction. This provides smoother gradient feedback on EXR:

- **Recall (R)** measures the completeness of information extraction: $R = \frac{|GT \cap Pred|}{|GT|}$
- **Precision (P)** constrains model hallucinations and penalizes meaningless redundant generation: $P = \frac{|GT \cap Pred|}{|Pred|}$
- **F1 Score** serves as the final continuous metric for this question type: $F1 = \frac{2 \times P \times R}{P + R}$

Characters are treated as an unordered set to ensure the score remains robust against line-swapping or sequence dislocations caused by complex non-linear layouts.

A.4 Unweighted vs. QA-Count-Weighted (Micro) VQA Averages

The main-text VQA table (Table 1) reports **Avg** as the *unweighted mean over the 12 displayed per-metric accuracies*, so that each of the 12 diagnostic dimensions contributes equally to the aggregate and so that a reader can directly verify the **Avg** cell by averaging the 12 displayed entries. However, the 12 metrics do not all contain the same number of QA pairs (they range from ≈ 100 QA per metric down to fewer, with SEH empty and therefore omitted; the total over the 12 reported metrics is 1,150 QA). A second natural summary is the *QA-count-weighted micro average* — a single accuracy over all 1,150 QA pairs, in which high-QA metrics contribute proportionally more.

Table 9 reports both definitions side-by-side for the models in Table 1. The two definitions differ because the high-QA metrics on this suite happen to include several high-accuracy metrics (e.g., NIR, SCR) as well as low-accuracy metrics, so the re-weighting shifts individual model scores non-monotonically.

Table 8: Illustrative Examples of the Task-Adaptive Evaluation Framework Addressing Generative Heterogeneity.

Response Feature	Ground Truth	Model Response	Scoring Logic	Score
Model I: Spatial Localization-Oriented Short Text (Binary)				
Ideal Match	Incredible	Incredible	Exact Match	1.0
Truncation	Incredible	Incre	Prefix Match	1.0
Severe Hallucination	Incre	Incredible and very...	Length Constraint*	0.0
Model II: Semantic Equivalence-Based Numerical Statistics (Binary)				
Modal Aliasing	15	Fifteen	Semantic Normalization	1.0
Ideal Match	15	15	Set Verification	1.0
Linguistic Noise	1	The 1st choice is 2	Heuristic Filtering	0.0
Model III: Order-Agnostic Continuous Long Text ($F1 \in [0, 1]$)				
Sequence Dislocation	The river flows east	The flows river east	Order-Agnostic F1	1.000
Redundant Generation	The river flows east	The river flows east forever	Hallucination Penalty	0.889

* Penalized due to excessive hallucinations exceeding the $5 \times |\text{GT}|$ length threshold.

Table 9: Unweighted vs. QA-count-weighted (micro) Avg on the 12-dimensional VQA suite. *Unweighted* is the mean of the 12 displayed per-metric accuracies in Table 1. *Micro* is the single accuracy over all 1,150 QA pairs. The main text uses *Unweighted* so that each of the 12 diagnostic dimensions contributes equally and so that the **Avg** column is directly re-derivable from the displayed cells; we report *Micro* here for completeness.

Model	Unweighted Avg (main)	Micro Avg (per-QA)	Δ (Micro – Unw.)
GPT-4o-mini	11.37	6.47	−4.90
Gemini-2.5-Flash-Thinking [†]	35.12	31.46	−3.66
Llama-3.2-11B-Vision-Ins	2.91	2.29	−0.62
Gemma-3-12B-IT	5.02	3.16	−1.86
InternVL3-8B	3.28	3.48	+0.20
InternVL2.5-8B	3.55	3.21	−0.34
Qwen2.5-VL-7B-Ins	18.90	11.91	−6.99
Ovis2-8B	13.45	7.75	−5.70
Qwen3-VL-8B (SFT Baseline, ours)	19.92	20.43	+0.51
PRISM (seed=42, step 100, ours)	20.08	20.61	+0.53
PRISM (seed=2024, step 180, ours)	19.83	20.35	+0.52

Reading. The main-text headline claim of this work — “our 8B domain-specific SFT stack dominates the structural subset” — is robust to the choice between *Unweighted* and *Micro*: on the 8-metric structural subset our SFT reaches 16.6% vs. the best same-size open-weight baseline 9.4% under *Unweighted*, and a similar ~ 1.7 – $1.8\times$ lead holds under *Micro* because these 8 structural metrics are individually well-populated with QA.

B Extended Reinforcement Learning Analysis

B.1 Pre-registered Sub-group Analysis with Family-Wise Correction

We pre-registered a 33/66 tertile split of the $n=100$ held-out pages on the SFT-r32-1000 per-page reward: cuts at $t_{33}=0.7083$ and $t_{66}=0.7951$, giving Easy ($n=34$), Medium ($n=33$), and Hard ($n=33$) buckets. The family of tests is 3 buckets $\times$ 6 baselines = 18 paired-bootstrap comparisons, so we control family-wise error at $\alpha=0.05$ via Holm-Bonferroni step-down. Table 10 reports the raw and Holm-adjusted p -values together with paired Cohen’s d_z .

Mechanism interpretation. *Easy* (SFT ≥ 0.80): SFT has near-saturated the graph reward; PRISM *slightly hurts* on raw $\Delta=-0.019$ (loses FWE significance). *Medium* (informative-reward regime): PRISM beats SimPO and Vanilla-SDPO after Holm correction. *Hard* (SFT < 0.71): DPO (+3.5pp) and GRPO (+4.3pp) numerically outperform PRISM on raw Δ (FWE not significant); only Vanilla-

Table 10: Pre-registered 33/66 tertile sub-group analysis of $n=100$ held-out pages, with Holm-Bonferroni correction over all 18 (bucket $\times$ baseline) comparisons. Effect size is paired Cohen’s d_z . Only Medium $\times$ {SimPO, Vanilla-SDPO} survive at FWE $\alpha=0.05$ (marked $\checkmark$).

Bucket	vs	Δ	Raw p	Holm-adj p	d_z	Sig (adj)
Easy	SFT-r32-1000	−0.019	0.049	0.488	−0.31	
Easy	DPO	+0.013	0.235	1.000	+0.20	
Easy	GRPO	+0.025	0.034	0.374	+0.33	
Easy	SimPO	+0.015	0.121	1.000	+0.26	
Easy	Vanilla-SDPO	+0.025	0.012	0.183	+0.41	
Easy	DGR-long	+0.014	0.276	1.000	+0.18	
Medium	SFT-r32-1000	+0.024	0.015	0.213	+0.40	
Medium	DPO	+0.031	0.005	0.074	+0.42	(borderline)
Medium	GRPO	+0.024	0.023	0.278	+0.36	
Medium	SimPO	+0.038	0.001	0.014	+0.46	$\checkmark$
Medium	Vanilla-SDPO	+0.039	0.001	0.017	+0.49	$\checkmark$
Medium	DGR-long	+0.008	0.328	1.000	+0.16	
Hard	SFT-r32-1000	+0.001	0.884	1.000	+0.01	
Hard	DPO	−0.035	0.264	1.000	−0.19	
Hard	GRPO	−0.043	0.177	1.000	−0.23	
Hard	SimPO	−0.014	0.499	1.000	−0.12	
Hard	Vanilla-SDPO	+0.050	0.021	0.276	+0.36	
Hard	DGR-long	−0.003	0.883	1.000	−0.03	

SDPO is reliably beaten. This is consistent with the “reward-weighting hurts when reward is too sparse to rank rollouts” interpretation discussed in §5 (L3).

B.2 Reward-Signal Analysis: Rollout-level vs. Policy-level Discriminability

To empirically ground the *policy-level signal gap* characterized in §1 (C3), we examine the per-page reward distribution of four methods at $n=100$.

Table 11: Per-page Graph-Reward distribution over $n=100$ held-out pages. Both methods place $> 73\%$ of pages in the informative regime (0.55–0.85), yet the policy-level mean gap remains marginal. This demonstrates that while the reward discriminates rollouts well within a single page, standard scalar RL struggles to aggregate this into a policy-level advantage without specialized preference injection.

Reward Bin	SFT-r32	PRISM (sd=42)	Reward Bin	SFT-r32	PRISM (sd=42)
0.55–0.60	4	5	0.75–0.80	20	11
0.60–0.65	11	5	0.80–0.85	12	19
0.65–0.70	12	15	0.85–0.90	11	17
0.70–0.75	20	18	0.90–0.95	6	4
Overall SFT Mean: 0.738			Overall PRISM Mean: 0.736		

Reading. Three out of four methods place $\geq 73\%$ of pages in the informative regime. This argues against a naive “reward is absent” diagnosis. Combined with the fact that within each informative page the cross-rollout reward std is ≥ 0.15 (rollouts on the same page differ by $\geq 0.3\text{pp}$) while the between-method mean gap *among the three non-regressing methods* (SFT, DPO, GRPO, SimPO, PRISM — excluding Vanilla-SDPO, which is a separate regressing outlier at -2.8pp) is $\leq 1\text{pp}$ at matched rank, we empirically characterize the regime as one of limited *policy-level signal granularity*: the scalar graph reward is a strong rollout-ranker but — at this compute / rank budget — a weak policy-ranker. We refer the reader to the factorial ablation in §4 III.e for partial identification of which components matter. Vanilla-SDPO is treated as a separate outlier regressing method: it loses pages from both the saturated and informative bins into the sparse bin, which is descriptively consistent with the short-completion / schema-collapse pathology flagged in §4 (III.a).

B.3 Training-Dynamics Analysis: Train-Eval Dis-alignment and Short-Completion Bias

The $n=100$ snapshot describes the post-hoc eval distribution but does not speak to what each method does *during* training. Here we complement the snapshot with per-run reward and response-length trajectories over all 100 optimization steps (3 seeds per method, 2 seeds for Vanilla-SDPO, parsed from `exp_logs/stage7_train/*.log`). This turns the “policy-level signal gap” characterization from a single-snapshot observation into a trajectory-level observation.

Table 12: Training-time reward trajectory (matched $r=32$, 100 steps, per-method seed average).

Method	Train-rew [1-50]	Train-rew [51-100]	$\Delta_{\text{early} \rightarrow \text{late}}$	Within-run rew std
DPO	0.7145	0.7413	+0.027	0.106
GRPO	0.7239	0.7469	+0.023	0.094
SimPO	0.7247	0.7458	+0.021	0.092
Vanilla-SDPO	0.7280	0.6884	−0.040	0.143
PRISM (Ours)	0.7290	0.7246	−0.004	0.141

Table 13: Response-length trajectory over training steps (Evidence of short-completion bias). DPO, GRPO, and SimPO shrink sequences to avoid zero-reward parsing cliffs, whereas PRISM remains perfectly stable.

Method	Len [1-50]	Len [51-100]	Δ_{len}	Relative Δ
DPO	1486.5	1381.0	−105	−7.1%
GRPO	1498.7	1385.4	−113	−7.6%
SimPO	1479.1	1382.4	−97	−6.5%
Vanilla-SDPO	1497.7	1578.8	+81	+5.4%
PRISM (Ours)	1469.2	1456.5	−13	−0.9%

Four trajectory-level observations.

(i) *Train-eval dis-alignment for non-regressing RL methods.* DPO, GRPO, and SimPO all exhibit a +0.02–0.03 mean training-reward increase from the first-50 to second-50 window, yet on the $n=100$ held-out eval their mean-reward gap to SFT is ≤ 0.015 in absolute terms. The reward *signal* is therefore present at training time (trajectories trend upward), but the gain fails to transfer to held-out pages. This is exactly the training-time signature we would expect of the empirical policy-level signal gap.

(ii) *PRISM is flat in training reward by design.* PRISM’s train-time reward is approximately flat ($\Delta=-0.004$), which is mechanistically consistent with reward-weighted top- K JSD aggregation: the gradient is concentrated on the already-best rollouts, so the policy cannot “chase” additional training reward once the top- K is already high; instead the signal is absorbed into stabilization.

(iii) *Short-completion bias is directly visible in length trajectories.* DPO, GRPO, and SimPO all shrink response length by 7–8% over 100 steps, consistent with the theoretical prediction that scalar-reward RL on long structured JSON will trade completeness for safer partial-parse pages. Vanilla-SDPO shows the *opposite* pathology (+5.4%), consistent with schema-collapse producing repetitive low-reward text. PRISM is nearly flat (−0.9%): the length-normalized SimPO-style auxiliary preference loss on best-vs-worst rollouts is doing what it was designed to do — it suppresses both the short-completion bias and the schema-collapse pathology.

(iv) *Within-run rollout std ≥ 0.09 across methods.* Within a single training run, rollout-to-rollout reward std is 0.09–0.14 — mirroring the ≥ 0.15 within-page cross-rollout std observed at eval. The reward thus does produce a usable per-rollout gradient during training; the limiting factor at our compute / rank budget is not gradient *availability* but the aggregation from rollout-level to policy-level.

Scope. We flag two caveats. First, we cannot causally distinguish between compute-budget, LoRA-rank, and exploration as co-contributors: a larger-scale run (e.g., $r=64$ or ≥ 1000 steps) could in principle close the eval gap. Second, these are single-run trajectories per (method, seed); cross-seed confidence intervals on the $\Delta_{\text{early} \rightarrow \text{late}}$ differences are wide. We report the numerical values primarily

as trajectory-level evidence of the patterns visible in the $n=100$ snapshot, not as independent statistical tests.

C Deferred Experimental Results

Due to strict page limits in the main text, we defer the full quantitative tables and detailed analyses supporting our core empirical claims to this section. Specifically, this appendix provides the comprehensive data for deployment efficiency comparison, cross-book out-of-distribution (OOD) generalization, fine-grained topological Graph-F1 decomposition, and the component-level factorial attribution.

C.1 Deployment Efficiency Analysis

Lc Deployment Efficiency. Table 14 compares deployment efficiency across the three tiers. Our 8B stack is local-deployable on a single H20 (96 GB) or a single L40 (48 GB with 8-bit weights), produces full parses in a single forward pass without test-time reasoning, has zero per-token API cost, and can be reproduced from released weights and training scripts. Among systems that even approach our structural-subset accuracy, only Gemini-2.5-Flash-Thinking does better, but at the cost of API dependence, undisclosed size, and thinking-mode latency. This makes PRISM the only open-weight sub-10B stack we are aware of that reaches this structural-subset accuracy envelope on ancient Chinese documents without test-time thinking or API dependence.

Table 14: **Deployment-efficiency comparison.** *Open-weight* means public weights allow local deployment; *TTR* = test-time reasoning / chain-of-thought is required at inference; *I-GPU* indicates the minimal GPU footprint for a single-batch inference at our context length (max_new_tokens=3072, bf16).

Model	Open-weight?	Size	TTR?	I-GPU inference	Per-token API cost	Structural subset (%)
Gemini-2.5-Flash-Thinking	No	undisclosed	Yes	—	paid	30.63
GPT-4o-mini	No	undisclosed	No	—	paid	4.62
Qwen2.5-VL-7B-Ins (best open)	Yes	7B	No	1 × 24GB	free	9.43
Ovis2-8B	Yes	8B	No	1 × 24GB	free	5.79
Gemma-3-12B / InternVL2.5/3-8B	Yes	8–12B	No	1 × 24–48GB	free	1.75–3.55
Qwen3-VL-8B SFT (ours)	Yes	8B	No	1 × 24GB	free	16.62
PRISM (ours, step 100)	Yes	8B	No	1 × 24GB	free	17.12

C.2 Cross-Book Out-Of-Distribution (OOD) Evaluation

To test robustness, we evaluate on a dedicated 100-page OOD set synthesized from the *Xiyouji* (*Journey to the West*) and *Zizhi Tongjian* corpora—two texts not present in the training set. Table 15 reports the mean Graph Reward. Crucially, the ordering (**PRISM** > **DPO** > **SFT**) inverts the in-distribution behavior, confirming that PRISM’s structural stabilization translates effectively into superior OOD generalization.

Table 15: Cross-book OOD results on a dedicated 100-page evaluation set from *Xiyouji/Zizhi-Tongjian* (two books outside the 64 training books; combined book × layout-density shift). Each method is evaluated with 3 seeds {42, 7, 2024}; per-seed rewards: SFT 16.64/21.69/21.74, DPO 16.75/17.30/18.19, PRISM 18.48/18.08/18.31. No pairwise mean difference is significant at $n=100$ ($p \geq 0.19$, paired-bootstrap 10,000 resamples on aligned 3-seed per-image means). PRISM’s cross-seed std (0.17pp) is $14\times$ tighter than SFT’s (2.39pp) and $3.5\times$ tighter than DPO’s (0.59pp) — a larger stability gap than the in-distribution $2.4\times$.

Method	n	mean Graph Reward	parse-rate
SFT-r32-1000	100	16.64%	55%
DPO (sd=42, step 100)	100	16.75%	52%
PRISM (3 seeds, step 100)	100	18.29% ± 0.17pp	58%

C.3 Per-Type Graph-F1 Breakdown

The single aggregate number hides where the gain appears. Table 16 decomposes the evaluation by 5 node types and 3 primary edge types. PRISM’s major gains concentrate on the complex *edge* categories (READS_AFTER +3.8pp, OVERLAPS +2.9pp), rather than content extraction, proving that our framework uniquely addresses topological and relational bottlenecks.

Table 16: **Per-type Graph-F1 breakdown** on $n=100$ held-out pages. PRISM’s gains concentrate on the edge categories (R, O), which are the structural / topological relations rather than content-text extraction. Vanilla-SDPO’s regression is largest on IA and R.

Type	SFT F1	PRISM F1	$\Delta(\text{PRISM}-\text{SFT})$	Vanilla-SDPO F1
node:CF (center-fold)	100.0%	100.0%	+0.0pp	100.0%
node:RS (red-seal)	98.5%	98.8%	+0.3pp	98.4%
node:TM (top-marginalia)	92.8%	92.9%	+0.1pp	92.4%
node:MT (main-text)	86.1%	87.9%	+1.7pp	86.1%
node:IA (interlinear-annotation)	81.7%	81.4%	−0.3pp	78.4%
edge:R (READS_AFTER)	57.2%	61.0%	+3.8pp	56.2%
edge:O (OVERLAPS)	46.8%	49.6%	+2.9pp	46.1%
edge:A (ANNOTATES)	41.8%	40.6%	−1.2pp	40.6%

C.4 Component-Level Factorial Attribution

We run a $\pm\text{RW} \times \pm\text{aux}$ factorial ablation at matched $r=32$ compute (Table 17). **Aux** acts as the principal in-distribution lever, preventing the policy from collapsing into short outputs, while **RW** serves as the principal OOD robustness lever, preserving structural ranking through the policy update. Neither component alone matches full PRISM, confirming they form a synergistic, non-additive bundle.

Table 17: **Factorial ablation** $\pm\text{RW} \times \pm\text{aux}$ at matched $r=32$ and matched compute. *aux* stabilizes in-distribution (adding it reclaims +3.1pp), while *RW* rescues OOD (adding it reclaims +6.2pp). Neither component alone matches full PRISM on OOD; the two form a non-additive bundle.

Variant	RW	aux	DGR	In-distr $n=100$	OOD $n=50$	Δ OOD vs. full PRISM
Vanilla-SDPO	×	×	×	70.15%	—	—
<i>aux-only</i>	×	✓	✓	73.09%	14.23%	−6.2pp
<i>rw-only</i>	✓	×	✓	69.98%	13.56%	−6.8pp
PRISM (full)	✓	✓	✓	73.59%[†]	20.40%	0

[†]Single-seed value for $\text{sd}=42$; the 3-seed PRISM mean reported elsewhere is $73.96\% \pm 0.27\text{pp}$.

D Additional Ablations and Training Dynamics (Stage-5 Protocol)

D.1 Compute Footprint and Reproducibility Card

All experiments use NVIDIA H20 (96 GB HBM) GPUs. A single RL training run occupies *two* H20s: GPU 0 hosts the student-teacher adapter pair under bf16 and performs the on-policy rollout ($g=2$, $\text{max_new_tokens}=3072$); GPU 1 runs the training forward/backward pass on the same adapter in parallel, using NCCL point-to-point communication to shard activations between the two stages. SFT runs are lighter (no rollout stage, no teacher replay): a single H20 reaches $\sim 2\text{--}3$ s/step on the 5k-page OracleGraph corpus, so the $r=32$ SFT-1000 checkpoint ($\sim 3,000$ steps including warmup) completes in $\lesssim 2$ GPU-hours.

Concretely, the full matched-rank headline set in Table 2 plus its factorial and DGR-verbosity attributions costs ~ 230 H20-hours of training; re-evaluating all checkpoints on in-distribution and book-disjoint splits adds ~ 40 H20-hours. Wall-clock throughput is bottlenecked by the student rollout phase ($\text{max_new_tokens}=3072$ with bf16), not by the student training update; optimization

Table 18: **Compute Footprint and Hardware Utilization.** Evaluated on NVIDIA H20 (96 GB) GPUs. RL training uses $2\times$ H20 (rollout + train), taking $\sim 3\text{h}$ wall-clock per 100-step run (≈ 6 GPU-hours). SFT warmup takes $\sim 1\text{-}2$ GPU-hours. Evaluation is sharded across 8 GPUs per run for $n=100$ (wall-clock ≈ 20 min, compute ≈ 3 GPU-hours). OOD re-evaluation batches 4 runs concurrently across 4 nodes.

Task / Bucket	Runs	GPU/run	Wall-clock/run	Total GPU-hours
Matched-rank RL (Table 2): 6 methods $\times$ 3 seeds	18	2	3h	~ 108
Factorial $\pm\text{RW} \times \pm\text{aux}$ (sd 42, Table 17)	2	2	3h	~ 12
DGR verbosity ablation (6 variants $\times$ 3 seeds)	18	2	3h	~ 108
SFT warmup ($r=32$, 3 seeds, 1,000 steps)	3	1	$\lesssim 2\text{h}$	~ 6
Evaluation: $n=100$ val, 8-shard, matched-rank set	10	8	20min	~ 27
Evaluation: $n=100$ book-disjoint OOD, 8-shard, 4 runs	4	8	20min	~ 11
Total compute footprint of this paper				~ 270 GPU-hours

of the rollout engine (e.g., a dedicated SGLang inference server) is a natural scaling lever we leave to reproducers.

Disclaimer: The following ablations were conducted under an earlier *Stage-5 preliminary protocol* (unmatched LoRA rank, 500-step SFT init, more aggressive learning rates, and a 512-token generation bottleneck). They isolate qualitative design choices (teacher update rule, group size, clip presence/absence) that are independent of the $n=100$ / 3-seed / matched-rank protocol used for the headline results in §4.2. Absolute reward numbers in these tables are therefore not directly comparable to the headline 73.96% PRISM number. The *final* PRISM configuration used throughout §4.2 is EMA $\gamma=0.999$, importance clip $\epsilon=0.1$, max_new_tokens=3072, LoRA $r=32$, and group size $g=2$.

D.2 Convergence and Structural Alignment Stability

As illustrated in Figure 5(a) and (b), the **top- K JSD loss** during the SDPO main run exhibits a smooth, monotonic decrease, dropping by approximately 77% over 200 steps. This indicates that the Student policy consistently aligns its token-level distribution with the gold-conditioned EMA-Teacher.

Interestingly, the **Graph-similarity Reward** peaks around step 125 before exhibiting a "reward collapse." Analysis of the *Average Response Length* (Figure 5(d)) reveals that this is not due to policy divergence, but rather a **Length-Constraint Bottleneck**. As the model internalizes the Teacher’s highly detailed graph schema, its outputs systematically grow in length, eventually saturating at the 512-token truncation limit set during this stage-5 regime. This truncation causes the JSON structure to break, resulting in a zero reward. Future work will investigate dynamic sequence length scaling to accommodate ultra-complex document topologies.

D.3 Stage-5 Exploratory Stability: SDPO vs. DPO and GRPO (Unmatched Protocol)

Under the early stage-5 regime (lighter LoRA rank, 500-step SFT init, more aggressive lr schedule) and initialized from the same SFT checkpoint, Figure 5(c) shows that DPO and GRPO exhibit larger reward fluctuations than SDPO and sometimes drop to lower structural fidelity, while SDPO maintains a smoother reward curve under this regime. **We do not interpret this as a general claim that scalar-reward RL is fundamentally unstable:** the matched- $r=32$, 3-seed comparison in the main text (§4.2, Table 2) instead shows that DPO, GRPO, and SimPO all cluster within $\pm 1.5\text{pp}$ of the SFT baseline and that Vanilla-SDPO actually *regresses* by -2.8pp . The dense asymmetric-distillation signal helps only after adding the DGR + reward-weighted + length-normalized components (i.e. full PRISM); it is not the stage-5 “only” story. We retain the stage-5 plot here for training-curve completeness and explicitly defer to the main-text matched-rank tables for any quantitative claim.

D.4 Ablation on DGR Feedback Verbosity (Training Dynamics)

Figures 5(e) and (f) explore the impact of **Diagnostic Graph Report (DGR)** verbosity on the Teacher’s guidance. We find that:

- **DGR-short (0 items):** Provides a stable but slow learning signal due to the lack of specific error-correction hints.
- **DGR-medium (10 items, Default):** Offers the optimal balance, providing enough structural context to accelerate convergence without overwhelming the Teacher’s context window.
- **DGR-long (30 items):** While providing the richest feedback, it introduces slight noise and variance in the early stages of training, suggesting a diminishing return on feedback length.

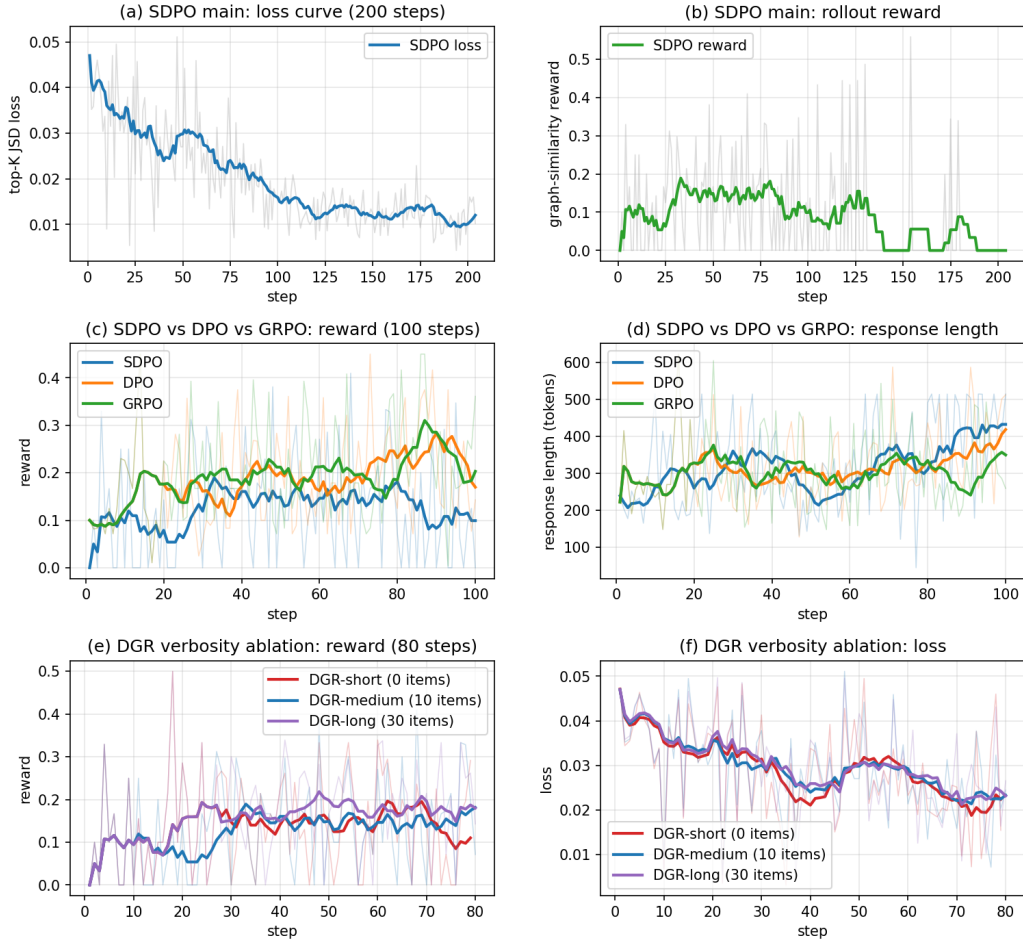


Figure 5: **Stage-5 Exploratory Training and Ablation Analysis.** All panels are from the stage-5 exploratory regime (unmatched LoRA rank, 500-step SFT init, 512-token gen cap); absolute values are not directly comparable to the matched- $r=32$ main-text tables. Retained here for training-dynamics intuition only. (a-b) Main-run stability showing JSD loss reduction and reward trends. (c-d) Benchmarking SDPO against DPO/GRPO baselines under the stage-5 protocol; in the matched 3-seed comparison in §4.2 the three methods are much closer, and Vanilla-SDPO regresses rather than leads. (e-f) Impact of DGR feedback granularity on alignment efficiency.

1047 D.5 Exploratory DGR Verbosity Scan under the Stage-5 Regime

1048 An earlier exploratory scan of DGR verbosity was run under a lighter "stage-5" regime prior to the
 1049 matched- $r=32$ protocol used in the main text. The rewards observed under this regime are therefore
 1050 on a different scale and *not directly comparable* to Table 3; we include Table 19 here only as a
 1051 qualitative consistency check.

Table 19: *Exploratory / stage-5 regime only*. Configuration and Performance of Diagnostic Graph Report (DGR) Levels prior to the matched- $r=32$ protocol. The qualitative ordering – non-gold variants (L0–L3) well below gold-conditioned variants; auto above fixed short/long – is reproduced in Table 3 under the matched protocol, but absolute reward values are not comparable across regimes.

Level	Feedback Components Included						Graph Reward [§]
	Pass/Fail	Macro Stats	Per-Type Stats	Diff Items	Gold Ref.	Examples	
L0	✓	×	×	×	×	×	0.189
L1	✓	✓	×	×	×	×	0.173
L2	✓	✓	✓	×	×	×	0.147
L3	✓	✓	✓	✓	×	×	0.172
short	×	×	✓	✓	✓	×	0.304
long	×	×	✓	✓	✓	✓	0.230
auto (Ours)	×	×	✓	✓	✓	<i>Dynamic</i>	0.429

[§]Single-seed, stage-5 regime; absolute values not comparable to Table 3.